ETHGLOBAL MUMBAI 2026 · IDEA DOCUMENT · VERSION 2

Wapsi

Recalls for robot software that no company can quietly switch off

If a piece of robot software is found to be dangerous, every robot running it slows down or stops by itself, and anyone can check any robot by scanning its QR plate. Recalls live on Arkiv, so no company can quietly edit or erase them.

Arkiv-centred | 0G Physical AI | World ID | Swarm

At a glance

What it is	A public recall network for robot software: AI models, policy checkpoints, firmware and planner prompts.
Who it's for	Robot makers, fleet operators, cities with robot permits, insurers and open-model hubs.
Core idea	Recalls live on Arkiv, where no company can quietly edit or erase them. Every robot checks before each task, trusts only records created by the right keys, and obeys by itself.
Prize picks	Arkiv (Build on Arkiv + Swarm bonus) · 0G (Physical AI) · World (Agent Pass and IDKit).
Build	Simulated robots (no hardware needed), TypeScript / Python, Arkiv SDK, 0G, World ID, Swarm.
Status	Final design, version 2. Includes every fix from two reviews (section 8). Ready to build.

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Wapsi means “return” in Hindi

1 The problem

Cars already get software recalls. When a defect is found, the maker publishes a recall and the cars get fixed. Robots and open AI models have nothing like this. When someone finds a dangerous flaw in a robot's model or firmware, thousands of robots around the world may still be running the same file, and nothing tells them to stop.

> **A real example: UniBLEed (2026)**
> In late August 2026, security researcher Olivier Laflamme disclosed flaws in Unitree's G1 humanoid robot: a Bluetooth flaw he named **UniBLEed**, which let someone nearby connect to the robot without confirmation or a password, plus two exploit chains assigned **CVE-2026-76639** (through the robot's chat_go service) and **CVE-2026-76640** (a buffer overflow in its Bluetooth server that ends with a command running as root). The attack was reportedly reproduced on four G1 robots and described as potentially wormable: a compromised robot could help spread it to other robots within Bluetooth range. Unitree confirmed the flaws and released patches. **But nothing makes sure every robot stops running the vulnerable version.**

Why today's way fails

Today	Why it fails
Recalls live on the maker's own servers	The maker may be slow, hacked, abroad or out of business, and it can quietly delete a recall about its own product.
Open-weight models often have no one running recalls	The publisher may be anonymous or gone, and nobody runs a recall server for a file anyone can download.
Different countries trust different authorities	Makers, regulators and researchers in different countries won't share one company's database.
Nobody can see who obeyed	A regulator can't tell which robots ignored a recall.

> **Regulation is arriving**
> On 15 September 2026, the San Mateo County Board of Supervisors (California) voted 4–0 for what County officials believe is a first-of-its-kind permit system for businesses using robots that operate at least partly without human control in places open to the public or to workers. To get a permit, a business must provide information about the robot's manufacturer, safety record and recalls. **Wapsi produces exactly that record, in public.**

2 The idea in plain words

Think of a vehicle recall notice pinned on a public board that nobody can quietly tear down. Every robot reads the board before each task. If its software is on the board, it slows down or parks itself. Anyone walking past can scan the robot's QR plate and see whether it is safe.

The whole story in five steps

1. A robot starts up and publishes exactly what software it is running (its **loadout**).
2. A problem is found. An AI scanner or a researcher raises an advisory, verified humans confirm it, and the maker or a regulator can issue a STOP.
3. Every robot running that software sees the recall within one check interval and obeys by itself.
4. Each robot posts a receipt saying it obeyed. Robots that stay silent show up on a public compliance board.
5. The maker publishes a fixed version. Robots that really run the fix go back to work; robots still on the old version stay stopped.

> **The 30-second pitch**
> "Cars get software recalls. Robots and open AI models get nothing. Wapsi is a recall network that no company can quietly switch off: recalls live on Arkiv, verified humans confirm them through World ID, and every robot checks before each task and obeys by itself."

3 How it works

Wapsi has three sides: the people who raise and confirm recalls (top), the robots that obey them (left), and the public who check robots (right). Arkiv sits in the middle as the shared record that nobody controls. Every reader checks **who created** each record before trusting it.

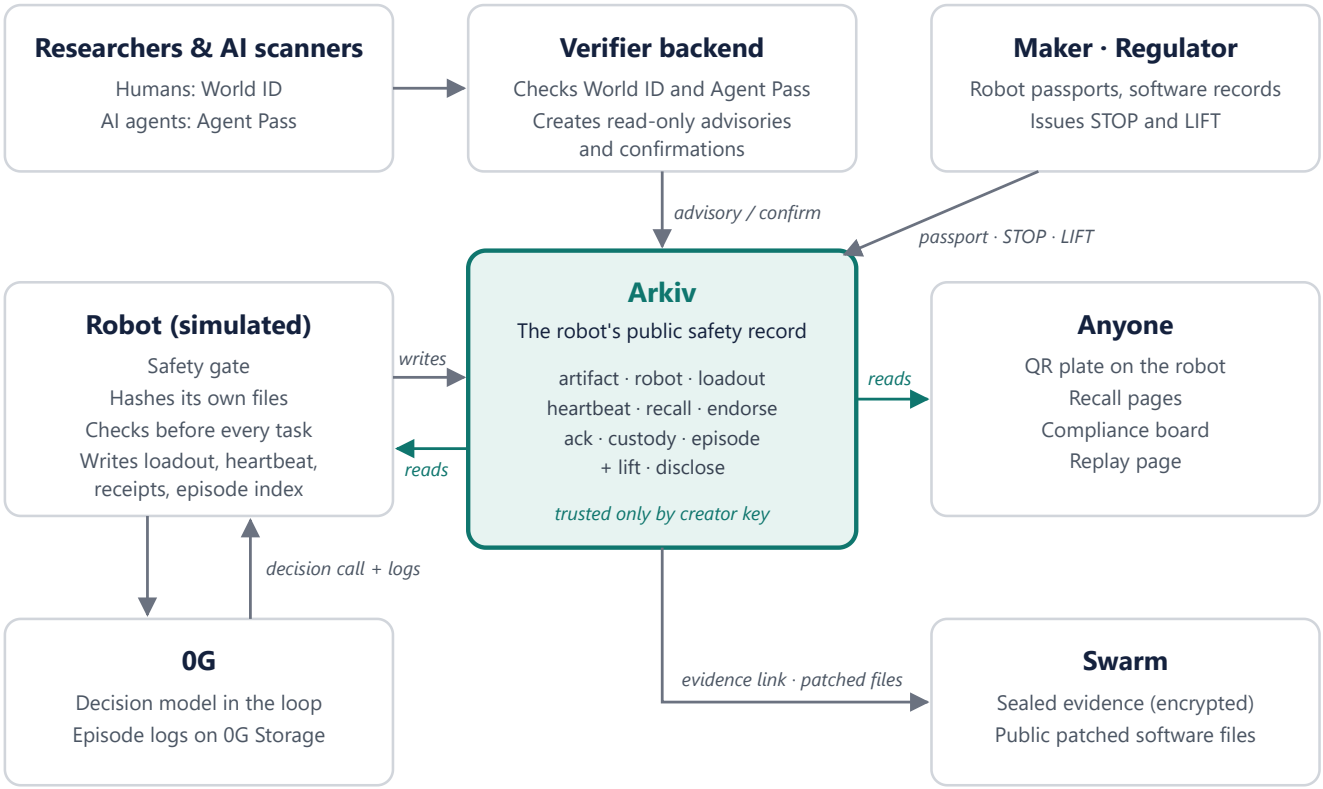


Figure 1. Who writes and who reads. Arkiv is the shared record in the middle; every reader checks the creator key of each record.

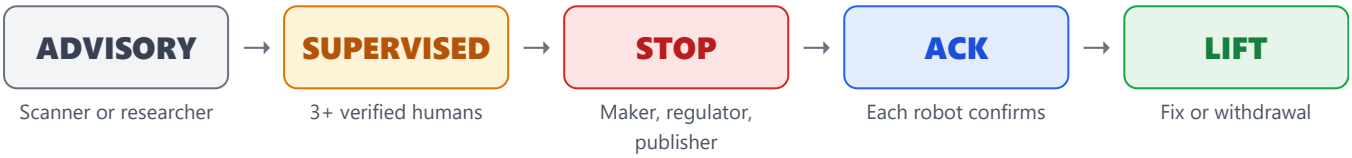
The parts

Part	Built with	Job
Safety gate (on the robot)	TypeScript or Python	Hashes everything the robot loaded, checks Arkiv before each task and every 30 seconds, trusts only records from the right keys, and slows or stops the robot.
Simulated robots	MuJoCo or a simple 2D simulator	Three robots doing simple tasks. No hardware needed.
Planner	0G model call	Turns a task into steps. Called inside the decision loop, with a request ID for every decision. Its prompt and model name are part of the loadout, so they can be recalled too.
Arkiv records	@arkiv-network/sdk	The robot's public safety record: 9 record types plus 2 event records (section 5).
Verifier backend	World ID: IDKit and Agent Pass	Checks real humans and AI scanners, then creates their advisory or confirmation as a read-only record.

Part	Built with	Job
Evidence and fixes	Swarm	Holds the encrypted reproduction until disclosure, and the public patched software files.
Episode logs	0G Storage	Every decision with its request ID. Indexed on Arkiv.
Public pages	Web app	QR plate, recall pages, compliance board, replay page.

> **Key principle**
> The gate trusts nothing it hasn't checked itself. It measures the files the robot really loaded, reads Arkiv directly, accepts a record only if the right key created it, and fails closed (slows down, then stops) if it can't read.

4 Recall levels



Level	Who can trigger it	What robots do	How it ends
ADVISORY	An AI scanner agent with a World Agent Pass, or a researcher verified with World ID. Both go through the verifier backend, which refuses a filing without a valid pass and then creates the record itself.	Nothing changes yet. The recall page shows the warning.	Ignored after 72 hours unless it reaches SUPERVISED or STOP. A regulator can also dismiss it.
SUPERVISED a state, not a separate record	Reached when at least 3 distinct verified humans confirm an advisory within the last 72 hours (one confirmation per person per recall) and confirmations outnumber disputes. Operators may set a higher threshold.	Slow speed, low force, tools put down safely, a human must be nearby.	Ends when fewer than the threshold of fresh confirmations remain. Only new confirmations keep it going. A regulator can dismiss it.
STOP	The maker, a regulator, or the software's publisher: keys each robot trusts.	Safe stop using a fixed routine, then posts an ACK.	Only a LIFT from the same key that issued the STOP, or from a regulator. A STOP that disappears is **not** a lift. Anyone can extend it, so it can't quietly lapse.
ACK	Each robot, with its own key.	Posts “I applied recall R” with R's content hash and what it did: stopped, slowed, left slow mode, or updated to H′.	Kept long. Feeds the compliance board.
LIFT	The key that issued the STOP, or a regulator.	**Fix published:** names the fixed version H′; robots that install H′ return to work, robots still on H stay stopped. **Withdrawn:** the STOP was a mistake; robots on H may resume.	—

> **Why the levels matter**
> If anyone could trigger STOP, a competitor could shut down your whole fleet. With levels, the community can only slow robots down, and only while fresh confirmations keep arriving. Only accountable keys (maker, regulator, publisher) can stop robots, and only the issuer or a regulator can lift a STOP.

5 Arkiv at the centre

Arkiv is a database chain. Each record (an "entity") has an owner wallet, a creator that never changes, an expiry, and attributes anyone can filter on. In Wapsi, Arkiv is the robot's public safety record: who made the software, what each robot is running, whether it is alive, which recalls apply, who confirmed them, whether the robot obeyed, and who is responsible for it today.

One rule above all: Arkiv's own docs warn that anyone can write a record with any attributes. So Wapsi never trusts a record because of its labels. It checks **who created it**.

5.1 The records

Record	What it holds	Trusted only if created by	Flags and expiry	What users see
wapsi.artifact	A model, adapter, policy, firmware or prompt: hash, source, publisher, "derived from" links	Its publisher or maker	Read-only. 180 days, renewed	Software page: who published it, and is it recalled?
wapsi.robot	Passport: robot public key, maker, model	A maker on the trusted list, so a fake robot can't register itself	Read-only. 180 days, renewed	QR plate header
wapsi.loadout	Exactly what the robot loaded, one attribute per piece, plus a sequence number	The robot key listed in its passport	Read-only. One per change. 180 days, renewed by the robot	"What is this robot running?" Also the starting point of the compliance board
wapsi.heartbeat	RUNNING / SUPERVISED / STOPPED, plus the loadout sequence number	The robot key	One per robot, updated and extended every minute. Expires after about 10 min (30 s in the demo)	QR plate. Expired means "unknown"
wapsi.recall	Software hash, level (ADVISORY or STOP), reason, severity, encrypted evidence link, creation block	ADVISORY: the verifier backend. STOP: a maker, regulator or publisher key	Read-only. ADVISORY: 72 h, extended by the verifier on each new confirmation. STOP: long, and anyone can extend	The safety gate, recall pages
wapsi.endorse	Confirm or dispute, the recall it refers to, a hashed human ID, creation block	The verifier backend, which keeps ownership	Read-only. 72 h	Confirmation count on the recall page
wapsi.ack	"I applied recall R", R's content hash, and the action: stopped, slowed, left slow mode, updated to H′	The robot key	Read-only. 180 days, renewed	Compliance board, robot history
wapsi.custody	Operator, contact, insurance	The robot's maker, who creates it at first sale. Its owner is the current operator	Editable by its owner. Ownership transferred on each resale. 180 days, renewed by the owner	"Who is responsible for this robot?"
wapsi.episode	Index of 0G Storage decision logs: request ID, software hashes, storage root	The robot key	Read-only. 30 days	Replay page

Two read-only event records complete the set: **wapsi.lift** (created by the key that issued the STOP, or a regulator; names the fixed version H′ or withdraws the STOP; a regulator's lift can also dismiss an advisory) and **wapsi.disclose** (the problem description, released at the disclosure deadline).

5.2 Every Arkiv feature changes something users see

Arkiv feature	How Wapsi uses it	What changes for users
Creator (never changes)	Every record is checked against the key that must have created it: the robot key from its passport, the trusted makers, regulators and publishers, or the verifier.	Fake heartbeats, fake receipts and fake recalls are ignored, even though anyone can write them.
Read-only flag	Set on recalls, confirmations, loadouts, receipts, lifts and episode indexes.	Nobody, not even the owner, can quietly rewrite them. Status changes are new records, so the full history stays visible.
Owner wallets	Only the owner can delete a record.	A maker can't touch a recall filed by a regulator or the verifier. If a maker deletes a STOP it filed itself, robots that saw it stay stopped and the board flags the deletion.
Expiry	Heartbeats, advisories, confirmations, the episode index.	"Alive" simply means the heartbeat exists, and old records clean themselves up. The 72-hour rule itself is enforced by the gate from the creation block, because owners can always extend their own records.
Anyone-can-extend	Set on STOP recalls when they are created.	Nobody can let a STOP quietly lapse. Affected robots extend it automatically.
Ownership transfer	Used only for wapsi.custody: the maker creates it, then it is transferred to each new operator, who keeps the contact details up to date.	A second-hand robot shows its new responsible operator. A fake custody record is ignored because the maker didn't create it. The robot's history (records made by the robot key) stays attached.
Attribute filters	Gate query, QR plate, compliance board.	One query answers "is there a STOP for anything I run?"

Checked against Arkiv's own docs

✓ The creator of a record never changes. ✓ The read-only flag blocks edits by anyone, but not deletion, transfer or extension by the owner. ✓ After an ownership transfer, the new owner can rewrite editable content, so Wapsi never transfers a record it relies on. ✓ Others can extend a record only if "permissionless extension" was set when it was created. ✓ Expiry emits no event. ✓ Queries return at most 100 results per page. ✓ The docs recommend filtering by owner, because anyone can write records with any attributes.

5.3 Example queries

```
-- Arkiv's current docs show SQL-like filters (=, AND). Exact syntax to confirm at the Arkiv booth.
-- Every result is also checked in the app: its creator must be the expected key.

-- Gate: STOP recalls on a file I run (one query per hash I load)
type = "wapsi.recall" AND level = "STOP" AND artifact = "0xPOLICY…"
--   keep only results created by a key I trust (maker, regulator, publisher)

-- Gate: confirmations for an advisory
type = "wapsi.endorse" AND recall = "0xRECALL…"
--   keep only records created by the verifier, inside the 72-hour window, one per human

-- QR plate: is robot R alive?
type = "wapsi.heartbeat" AND robot = "0xROBOT…"
--   keep only if created by R's key; no result means "unknown"

-- Compliance board
--   1. each registered robot's latest loadout:   type = "wapsi.loadout" AND robot = "0xROBOT…"
--   2. receipts for the recall:                  type = "wapsi.ack" AND recall = "0xRECALL…"
--   3. in the app: robots whose latest loadout has the recalled hash and no receipt,
--      shown as "silent" (no live heartbeat) or "running without a receipt" (alive)
```

Results come in pages of up to 100; read every page by paging on the record key. If the creator can be used inside a query, add it there too.

5.4 What stays off Arkiv, and why

Kept off	Where it goes	Why
Decision logs and video	0G Storage. Arkiv keeps only the index	Too large for Arkiv.
Evidence bundles	Swarm, encrypted	Large, and sensitive until disclosure.
The motor control loop	On the robot	Needs millisecond timing. Arkiv is checked per task, not per motor tick.
Exact location	Nowhere public	It would reveal when homes are empty. City level only.

5.5 Why not a normal database?

- A maker could delete or rewrite recalls about its own products.
- Makers, regulators and researchers in different countries won't trust one company's database.
- Open-weight models often have no one running a recall server at all.
- Expiry, ownership and permissions would have to be rebuilt by hand, and everyone would have to trust whoever runs the database.

6 Step-by-step flows

A. Start-up and every task (the safety gate)

1. The robot hashes everything it loaded (policy, adapter, firmware, planner prompt, 0G model name) and writes **wapsi.loadout**.
2. Before each task, and every 30 seconds (2 seconds in the demo), it asks Arkiv for recalls matching its hashes and keeps only records created by the right keys.
3. STOP from a trusted key: safe stop with a fixed routine, then **wapsi.ack**.
4. An advisory with at least 3 fresh, verified confirmations: slow speed, low force, tools put down safely, then **wapsi.ack**.
5. Nothing found: run the task.
6. Arkiv can't be read: 5 minutes' grace, then supervised mode, then stop. It fails closed, but gently.
7. The heartbeat is updated every 60 seconds.

B. Recall lifecycle

ADVISORY → SUPERVISED → STOP → ACK → LIFT, as described in section 4. Every step leaves new read-only records on Arkiv (advisory, confirmations, STOP, receipts, lift), so the whole history of a recall stays visible.

C. The QR plate (for bystanders, cities and insurers)

Scan the plate on any robot to see:

- its passport (maker and model);
- its current loadout;
- its live heartbeat (or “unknown” if it went silent);
- any open recalls and how many people confirmed them;
- the current responsible operator.

The plate shows only records created by the robot's registered key, the trusted makers and regulators, or the verifier, so nobody can paint a fake green status.

The status colour is simple: **green** running, **amber** supervised, **red** stopped, **grey** unknown.

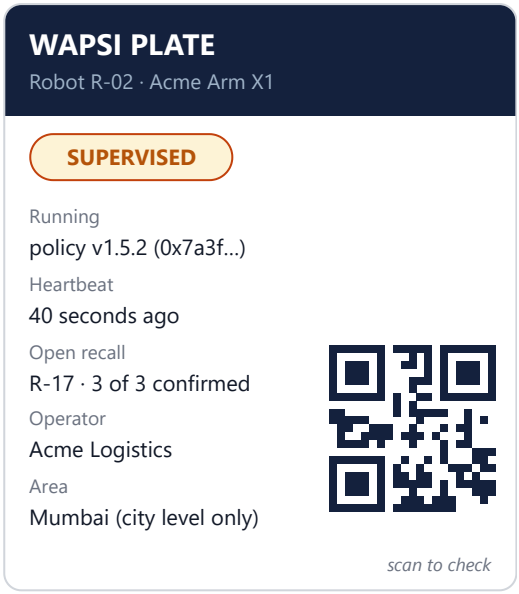


Figure 2. Example QR plate (illustration).

D. Sealed evidence, in two stages (Swarm bonus)

1. The researcher uploads the reproduction (video, steps, script) to Swarm, encrypted.
2. The Swarm reference, which contains its own decryption key, is itself encrypted to the maker's and the regulator's public keys before it goes on the recall record. No readable link ever goes on a public record.
3. At the disclosure deadline, a **wapsi.disclose** record publishes the **description** of the problem.
4. The working reproduction is released only after a fix is out (a LIFT exists) or with the regulator's approval, so people near unpatched robots aren't put at risk.
5. Patched software files go on Swarm unencrypted, so anyone can download the fix and check its hash.

E. Resale

The current operator transfers **wapsi.custody** to the buyer. Its creator stays the maker, so a custody record made by anyone else is ignored. The QR plate shows the new responsible operator, and the robot's recall history (records created by the robot key) stays attached.

F. Replay

The replay page reads **wapsi.episode** from Arkiv and opens the 0G Storage logs: “show every decision this robot made with recalled software”.

G. If a maker deletes its own STOP

Robots that saw the STOP stay stopped, because a STOP that disappears is not a lift. Their receipts carry the STOP's content hash, so the compliance board shows “STOP deleted by its issuer”. A regulator can issue its own STOP, which the maker cannot delete.

7 Safety gate logic (sketch)

```
const STOP_KEYS  = new Set([MAKER, REGULATOR, PUBLISHER]); // may issue STOP
const REGULATORS = new Set([REGULATOR]);                   // may lift any STOP or dismiss advisories
const VERIFIER   = "0xVERIFIER…";                          // creates advisories and confirmations
const WINDOW     = blocksIn(72 * HOURS);                   // 72 h measured in Arkiv blocks
const THRESHOLD  = 3;                                      // operators may raise it

async function checkBeforeTask(): Promise<Mode> {
  const hashes = measureLoadedFiles();                // the gate hashes the files itself
  const read   = await readArkiv(hashes);             // recalls, confirmations, lifts, current block
  if (!read.ok) return graceThenSupervisedThenStop(); // fail closed, gently

  // STOP: remembered locally, so a STOP that disappears is NOT a lift
  for (const r of read.stops) if (STOP_KEYS.has(r.creator)) remember(r);
  const stop = remembered().find(r =>
      hashes.includes(r.artifact) && !withdrawn(r, read.lifts));
  if (stop) { await safeStop(); await writeAck(stop, "STOPPED"); return "STOPPED"; }

  // SUPERVISED: a state reached by fresh, verified confirmations
  const sup = read.advisories.find(a =>
      a.creator === VERIFIER && !dismissed(a, read.lifts) &&
      freshConfirmations(a, read.endorsements, read.block) >= THRESHOLD);
  if (sup) {
    limit({ speed: "SLOW", force: "LOW", tools: "PUT_DOWN" });
    await writeAck(sup, "SLOWED");                    // "LEFT_SLOW_MODE" is written when it ends
    return "SUPERVISED";
  }
  return "RUNNING";
}

// Only confirmations the verifier created, inside the window, one per human,
// and only while confirmations outnumber disputes.
function freshConfirmations(a, ends, now) {
  const fresh = ends.filter(e => e.creator === VERIFIER && e.recall === a.key &&
                                 now - e.createdBlock <= WINDOW);
  const yes = new Set(fresh.filter(e => e.vote === "CONFIRM").map(e => e.human));
  const no  = new Set(fresh.filter(e => e.vote === "DISPUTE").map(e => e.human));
  return yes.size > no.size ? yes.size : 0;
}

// A STOP is withdrawn only by its own issuer or a regulator.
function withdrawn(r, lifts) {
  return lifts.some(l => l.recall === r.key && l.kind === "WITHDRAWN" &&
                         (l.creator === r.creator || REGULATORS.has(l.creator)));
}
// An advisory is dismissed only by a regulator.
function dismissed(a, lifts) {
  return lifts.some(l => l.recall === a.key && l.kind === "DISMISSED" &&
                         REGULATORS.has(l.creator));
}
```

- **Every record is checked against the key that must have created it.** Labels alone mean nothing.
- **A STOP applies only while the robot runs the recalled hash.** A robot that installs the fixed version H′ no longer matches it, so it returns to work; a robot still on H stays stopped.
- **safeStop** is a fixed routine (halt, put the tool down, go home). The AI never decides how to stop.
- **The robot installs a fix only if** its wapsi.artifact record was created by the robot's maker or publisher, and the downloaded file matches that hash.
- **The robot writes its own receipts** when it enters and leaves slow mode, so the history survives after an advisory expires.

- **createdBlock** is a read-only attribute the verifier writes when it creates the record (or Arkiv's own creation block, if reads expose it).

8 Fixes applied after review

This version includes the four fixes from the first review and seven more from a second review against Arkiv's documentation.

Fix	Problem	What we do
1. Counting real humans	One person can make ten wallets, so wallets can't be counted as humans. And if a confirmation were handed over to the person, Arkiv lets a new owner rewrite an editable record while its creator still names the verifier.	The backend checks World ID first (World's rules require backend validation anyway), then creates the confirmation as a **read-only** record and keeps it. The robot counts only verifier-created confirmations, one per hashed human ID per recall. Like an office stamp on a form that stays in the office file.
2. Fake records	Anyone can write an Arkiv record with any labels: a fake green heartbeat, a fake receipt, a fake recall.	Every record is accepted only if the right key created it: the robot key from its maker-created passport, a trusted maker, regulator or publisher, or the verifier.
3. The 72-hour rule	Owners can always extend their own records, so expiry alone can't make an advisory lapse.	The gate counts only confirmations created in the last 72 hours, measured in Arkiv blocks. Expiry is just cleanup.
4. Supervised mode never triggering	The gate looked for recalls labelled “SUPERVISED”, but advisories are filed as ADVISORY and never edited.	SUPERVISED is a state: any advisory with enough fresh confirmations.
5. Silent robots missing from the board	An offline robot's heartbeat expires, so a board built from heartbeats never lists it.	The board starts from each registered robot's latest loadout and uses the heartbeat only to mark it alive or silent.
6. A maker deleting its own STOP	Read-only stops edits, not deletion by the owner.	Robots remember STOPs, receipts carry the STOP's content hash so the deletion shows on the board, and regulators can issue their own STOP.
7. Who can lift a STOP	Any trusted key could lift, so a maker could lift a regulator's STOP.	Only the STOP's issuer or a regulator can lift it. Two kinds of lift: fix published (H′) or withdrawn.
8. Provisional history	When an advisory lapses, its history could disappear with it.	The robot writes its own receipts: “entered slow mode because of R” and “left it because R lapsed”.
9. Releasing evidence safely	Disclosure depended on the researcher remembering to publish the key, and publishing a working attack while many robots are unpatched endangers people near them.	Two-stage disclosure (flow D): the description at the deadline, the working reproduction only after the fix or with regulator approval. Optionally, time-lock encryption so the description opens by itself.
10. Being honest about hacked robots	A robot that is already hacked can lie about what it runs.	Say it plainly: Wapsi protects robots that aren't hacked yet. A hacked robot shows up as silent, or as never confirming, on the compliance board. In production the gate runs outside the AI's reach.
11. Recall arrival time	“2–4 seconds” contradicted the 30-second checks.	A recall arrives within one check interval plus about one Arkiv block. Quote the time you measure.

Smaller fixes

- Queries use the SQL-like style in Arkiv's current docs, and the board pages through results (at most 100 per page).
- The UniBLEed example now separates the Bluetooth flaw from the two CVE-numbered chains.
- The demo says “put tools down safely”, not “drop tools”, and the roles give three people a phone for the three confirmations.

- wapsi.loadout has a defined expiry (180 days, renewed), and patched files are public on Swarm (the openable link the Swarm bonus asks for).
- Wapsi is pitched as recalls for **any** robot-software safety defect, not only backdoors: a practitioner review found no publicly confirmed case of deliberately backdoored weights in a deployed production model as of September 2026, while real defects like UniBLEed already exist.

9 Honest limitations

Level	Limitation	How we handle it
HIGH	A robot that's already hacked can lie about what it runs.	Wapsi protects robots that aren't hacked yet; silent robots show on the compliance board. Production needs the gate in a secure chip, outside the AI's reach.
MEDIUM	The verifier backend is trusted to say “this is a real, unique human” and “this agent has a valid pass”.	World requires backend validation anyway. The backend can only cause slow mode, never STOP. Ask World whether proofs can be published for anyone to re-check.
MEDIUM	Griefing: three colluding people, or people renting out their World IDs, could slow a whole fleet.	Slow mode only, never STOP; it lasts only while fresh confirmations keep arriving; operators can raise the threshold; the verifier limits how many recalls one person can confirm per day; a regulator can dismiss an advisory.
MEDIUM	Recalls match exact hashes, so a copy with one byte changed escapes a recall.	wapsi.artifact carries “derived from” links, and robots load only software whose record was created by their maker or publisher.
MEDIUM	A maker can delete a STOP it issued itself.	Robots that saw it stay stopped, the board flags the deletion, and regulators can issue their own STOP.
MEDIUM	Robots trust the Arkiv server (RPC) they read from.	Read from two endpoints, or run our own node.
MEDIUM	Makers have to build the gate into their robots.	Start with open-model hubs and fleet operators; permits like San Mateo County's create the push.
LOW	If an open model's publisher is gone, nobody can STOP it.	The community can still put robots into slow mode, and a regulator can issue a STOP.
LOW	Heartbeats reveal when a robot is working.	Status only, city-level area, and operators can choose a longer interval.
LOW	Arkiv records expire unless extended.	STOP recalls are created with anyone-can-extend, and affected robots extend them. Confirm the maximum lifetime at the Arkiv booth.
LOW	Arkiv sends no “expired” event.	Robots poll every 30 seconds.
LOW	One heartbeat update a minute per robot.	Fine for the demo; real fleets would batch.
LOW	Each robot needs a funded testnet wallet, and the testnet may reset.	Keep a re-seed script ready.
LOW	“Remove the blockchain” test is about 7.5 / 10.	Real multi-party trust, but it doesn't protect money directly. Say so if asked.

10 Sponsors and requirements

ETHGlobal allows up to three partner picks. If a partner has several tracks, you can be eligible for all of them while counting as one pick.

Pick	Prize	They require	How Wapsi meets it
Arkiv Build on Arkiv (from scratch)	$3,000 1st · $1,000 2nd + $500 Arkiv + Swarm bonus (doesn't use a pick)	Built from scratch at Mumbai. Writes to Arkiv's current testnet (SDK or JSON-RPC), and at least one user-facing feature reads from it; a write-only log doesn't count. Public open-source repo; say how AI was used. README with writer wallet addresses and one runnable query. /arkiv/schema.md and /arkiv/friction.md. Judged 25% each on friction, reach, why Arkiv, and how you use Arkiv. One Arkiv track prize per team.	9 record types plus 2 event records, all read by public pages. The gate, QR plate and compliance board all read Arkiv.
0G Physical AI: Agents with a Body	$2,500 1st · $1,500 2nd + $1,000 best build on real hardware	A 0G call in the decision loop. A per-decision episode log on 0G Storage. A replayable simulation or a live hardware demo. A safe-stop fallback; a fallback that fakes decisions disqualifies. Public repo with README, demo under 3 minutes, live link, contract addresses (if any).	The planner calls 0G for every decision, with a request ID. Logs go to 0G Storage and are indexed on Arkiv. With 0G off, the robot pauses safely.
World Agent Pass + IDKit	Agent Pass: $2,500 1st · $1,500 2nd · $1,000 3rd IDKit: $1,250 each for up to 2 teams	Agent Pass: integrate the official Agent Pass; show the full journey, a denied, expired or cancelled path, and validate results in a secure backend; include an integration debrief. IDKit: a working app; a credential verified server-side or on-chain; explain why it is the minimum sufficient assurance; a success path and an alternative path; a debrief.	AI scanners file only through the backend with a valid Agent Pass, and are refused without one. Humans confirm with Proof of Human, one confirmation per person per recall, because uniqueness is all Wapsi needs to know.

Arkiv files checklist

File	What goes in it
/arkiv/schema.md	All records: creator, owner, flags, expiry and the fields we query.
/arkiv/friction.md	What got in our way, with reproducible bug reports.
/arkiv/reach.md	Who uses it: makers, fleet operators, cities with permit rules, insurers, open-model hubs such as LeRobot. Whether they can try it today, and whether we keep building after Mumbai.
/arkiv/why-arkiv.md	What users would lose with Postgres, and what we kept off Arkiv and why (section 5.4).
/arkiv/how-you-use-arkiv.md	Filters, creators, read-only records, expiry and transfer, and what each one changes for users (section 5.2).
/arkiv/swarm.md	What's on Swarm vs Arkiv and which field holds the reference; one entity key with its Swarm reference and an openable link (use a public patched file, not the sealed evidence); the postage batch ID and who bought it; the Bee node or service; the files that upload and read; steps to run one upload and one read; the demo timestamp; obstacles.
README	Writer wallet addresses, one runnable query, and how AI was used.

Backup plan
If 0G integration gets stuck, cut 0G and switch the third pick to the Ethereum Foundation's Best CROPS App ($2,000 for 1st). It needs a README section titled CROPS, a demo of 3 minutes or less showing the lead property, and no third-party trackers in the client. The story becomes: "the recall still works even if the maker's server is switched off."

11 Demo plan

Finalists get 7 minutes: a 4-minute demo, then 3 minutes of questions. Judges score technicality, originality, practicality, usability and WOW factor.

The story
Three simulated robots run software v1.5.2. A safety defect is found (modelled on a real case such as UniBLEed; no hacking is shown). The robots are recalled, obey, and are fixed with v1.5.3.

The 4-minute script

Time	What judges see	What you say
0:00–0:20	Slide: the UniBLEed headline, and "San Mateo County now asks for robot recalls".	"Cars get software recalls. Robots and open AI models get nothing. Wapsi is that recall system, and no company can quietly switch it off."
0:20–0:40	Three robots working in the simulator. Scan one QR plate on a phone: passport, software, green heartbeat, no recalls.	"Anyone can check any robot. This lives on Arkiv, not on our server."
0:40–1:05	An AI scanner files a warning **without** an Agent Pass: refused. **With** the pass: accepted. Robots keep working.	"AI can raise an alarm, but only with a real human behind it."
1:05–1:50	Three teammates confirm with World ID on their phones. The counter goes 1/3, 2/3, 3/3; robots slow down and put their tools down safely. A big timer shows the measured time. One teammate tries to confirm twice: refused.	"One human, one vote. The community can slow robots down, but can't shut them down."
1:50–2:30	The maker presses STOP. Robots do their safe stop and post receipts. Robot 3 is offline: its plate turns grey, and the board shows "2 of 3 obeyed, 1 silent".	"Now regulators can see who ignored a recall."
2:30–2:55	The maker publishes the fix v1.5.3 (a LIFT). Robots 1 and 2 install it and resume. A robot still on v1.5.2 stays stopped.	"The recall lifts only when the robot really runs the fixed software."
2:55–3:20	Replay page: every decision made with the bad software (Arkiv index + 0G logs with request IDs). Then 0G is switched off and the robot pauses safely.	"Every decision is on record, and if the AI brain goes offline, the robot stops safely."
3:20–3:30	Closing slide.	"Wapsi: recalls for robots that no company can quietly switch off."

That leaves about 30 seconds of spare time, which you will need.

Keep ready for questions (not in the 4 minutes)

- **Fake record:** an outsider posts a fake green heartbeat for robot 3; the plate ignores it because the robot's key didn't create it.
- **Resale:** transfer the robot; the plate shows the new operator with the recall history still attached.
- **Internet cut:** grace period, then slow mode, then stop.
- **Sealed evidence:** a Swarm reference that can't be opened without the key.
- **The Arkiv explorer** showing the real records.

Setup checklist

1. **Shorten timers for the demo, and say so out loud:** 2-second checks instead of 30, a 30-second heartbeat expiry instead of 10 minutes, a 20-second grace period instead of 5 minutes.
2. **Screen layout:** robots on the left, QR plate and compliance board on the right. Show the phone through a webcam or screen mirroring.
3. **Phones:** three teammates' phones with World test identities ready. Don't rely on judges.
4. **Wallets:** fund every robot wallet the night before. Keep a re-seed script for testnet resets.
5. **Backup video:** record the full demo once it works, and switch to it if anything breaks. Keep a version under 3 minutes for the submission and the sponsor judges.
6. **Rehearse five times with a timer.** Cut whatever runs over.

Role	During the demo
Speaker	Tells the story and keeps eye contact with judges. Makes confirmation 1 on their phone.
Driver	Runs the laptop: simulator, console, pages.
Phone A	Makes confirmation 2, then tries to confirm a second time (refused).
Phone B	Makes confirmation 3. Watches the clock and holds the backup video.

What each sponsor must see

Sponsor	Must show
Arkiv	Real records in the explorer, one runnable query in the README, the /arkiv files, and a Swarm reference for the bonus.
0G	A 0G call in each decision, request IDs in the logs, and a safe stop when 0G is off.
World	Agent Pass refused, then accepted. IDKit: one success and one refused second confirmation. The debrief notes.

12 Tough judge questions

Question	Short answer
“Why not a normal database?”	A maker could delete or rewrite recalls about its own products, and countries won't trust one company's server. Open models often have no one to run one.
“Can't anyone write fake records to Arkiv?”	Yes, anyone can write records. That's why the gate ignores labels and checks which key created each record.
“Can a hacked robot lie?”	Yes. Wapsi protects robots that aren't hacked yet, and silent robots show up on the compliance board. In production the gate sits in a secure chip.
“Could a competitor shut down my fleet?”	No. The community can only slow robots down, and only while three or more fresh verified confirmations keep arriving (operators can require more). Only the maker, a regulator or the publisher can STOP.
“What if the maker deletes its own STOP?”	Robots that saw it stay stopped, the board shows the deletion, and a regulator can issue a STOP the maker can't delete.
“What stops recall spam?”	AI scanners need an Agent Pass, humans count once each through World ID, the verifier rate-limits each person, and unconfirmed advisories are ignored after 72 hours.
“How fast does a recall arrive?”	Within one check interval plus about one Arkiv block: under 30 seconds plus a block in real use, about 2 seconds plus a block in the demo. Quote the time you measured.
“Does it work on real robots?”	It runs in simulation today and is designed to plug into robot software stacks such as LeRobot. Real robots would add a secure chip for the gate.

Question	Short answer
“Why blockchain at all?”	Many writers who don't trust each other, across countries; nobody can quietly edit or delete; one-human-one-vote filing; records anyone can read without asking us.
“What about privacy?”	Location is city level only, evidence is encrypted, and confirmations carry only a hashed ID, never personal data.

13 Build plan

Scope

Priority	Items
MUST	Records: artifact, robot, loadout, heartbeat, recall, endorse, ack, lift, with read-only flags and creator checks. Safety gate with STOP, SUPERVISED (72-hour window) and fail-closed. Three simulated robots. Verifier backend: World ID confirmations and the Agent Pass scanner (with refusal). 0G call in the decision loop, logs, and safe stop when 0G is off. QR plate and compliance board (including silent robots). LIFT rules (issuer or regulator).
SHOULD	Custody transfer (resale). Episode index and replay page. Swarm sealed evidence with two-stage disclosure and public patched files. Regulator dismissal of advisories.
STRETCH	Real hardware at the 0G booth. Reading from two Arkiv endpoints. Time-lock encryption for disclosure.

Team split (four people)

Person	Owns
Robot & gate	Simulated robots, the safety gate, the safe-stop routine, the 0G planner and logs.
Arkiv & data	Record schema and flags, SDK writes and queries, creator checks, extension and ownership transfer, re-seed script, Swarm.
Identity & console	World ID verifier backend, Agent Pass scanner, maker/regulator console (STOP, LIFT, dismiss).
Frontend & pitch	QR plate, recall page, compliance board, replay page, slides, backup video.

Time blocks (hours from when you start building)

Hours	Goal of the block
0–5	Arkiv schema with flags. Writes for robot, artifact, loadout and heartbeat, with creator checks. Three simulated robots moving. Gate skeleton.
5–10	Gate reads recalls; STOP routine, local memory of STOPs, and receipts with the STOP's content hash; fail-closed grace. 0G call per decision with request IDs; logs to 0G Storage; safe stop when 0G is off.
10–15	Verifier backend: World ID confirmations (read-only, one per human per recall) and the Agent Pass scanner with the refusal path. Supervised mode with the 72-hour window.
15–20	QR plate, recall page, compliance board (from loadouts, with silent robots). LIFT rules. Custody transfer.
20–25	Swarm sealed evidence, two-stage disclosure and public patched files. Replay page. Demo timers, re-seed script, first full run-through.
25–30	/arkiv files, swarm.md and README. Record the backup video. Five timed rehearsals.

If the 5–10 block runs late, cut 0G and use the EF CROPS backup.

Questions to ask at the booths on day 1

- **Arkiv:** can read-only and anyone-can-extend be set together on one record; can the creator be used inside queries; are OR and IN supported; do reads expose a creation block; write limits; testnet resets; maximum lifetime.
- **0G:** can it serve our own adapter, or must the decision model be theirs? How do we get a request ID for each call?
- **World:** Agent Pass for an AI agent that files recalls (docs at auth.world.org/docs); can proofs be re-checked by others; test identities for three phones; rate-limiting one person per action.
- **Swarm:** how to get a postage stamp batch at the event.

14 Sources

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- **San Mateo County robot permit system (15 September 2026, 4–0, manufacturer, safety record and recalls)** — County of San Mateo
  https://www.smcgov.org/ceo/news/san-mateo-county-supervisors-pass-new-public-safety-worker-protections-autonomous-robots
- **Arkiv entity lifecycle: read-only flag, ownership transfer, extension by others, creator never changes, no expiry events** — Arkiv (GitHub)
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- **Arkiv querying: SQL-like filters, at most 100 results per page, filter by owner because anyone can write any attributes** — Arkiv docs
  https://docs.arkiv.network/json-rpc/querying-data/
- **Up to three partner picks; finalist format (4-minute demo, 3 minutes of questions) and judging criteria** — ETHGlobal Mumbai event details
  https://ethglobal.com/events/mumbai/info/details
- **Prize amounts and track requirements (Arkiv, 0G, World, Ethereum Foundation)** — ETHGlobal Mumbai prizes
  https://ethglobal.com/events/mumbai/prizes
- **World ID human-in-the-loop and Agent Pass integration** — World Developer Docs
  https://docs.world.org/agents/human-in-the-loop/integrate
- **No publicly confirmed deliberately backdoored weights in a deployed production model (as of September 2026)** — securing.ai practitioner review
  https://securing.ai/backdoor-attacks-ml/

Facts were checked on 6 November 2026. Exact SDK and query syntax should be confirmed at the sponsor booths before building.